

AI-Powered Eldercare Platform

PulseCare

AI-Powered Senior Health Monitoring

February Jiang · yj2918

"Your body has a story. Pulse remembers it."



Motivation & Positioning

72M seniors hide pain from families — crises arrive without warning.

Five years ago, I lost my grandmother.

She hid her physical pain to avoid worrying our family. That silence — and the fall that followed — is why PulseCare exists.

Seniors Under-Report

72M independent US seniors systematically hide symptoms to avoid burdening family. Disclosure only happens when pain is severe enough to admit — often too late.

Private Family Doctor

Our Positioning: An always-on "private doctor" for the family, providing early warnings and actionable, professional medical guidance.

Shared Family Bond

Daily conversational check-ins extract structured health signals invisibly. Caregivers see synthesis, not noise — ranked alerts + predictive 48-hour insight.

Business Model & Technical Building

High-margin SaaS powered by conditional multi-agent AI — cost scales sub-linearly with users.

Unit Economics

98.8%

Gross Margin

\$29.00 /mo

Subscription per caregiver

~\$0.06

LLM cost (Gemini 2.0 Flash, conditional routing)

~\$0.36

Total COGS (infra + compute)

\$28.64

Contribution margin per user

Moat: after 60 days the senior's longitudinal baseline is the product — irreplaceable clinical memory.

Conditional Multi-Agent Routing

LangGraph: Companion Agent handles every check-in cheaply (~\$0.001). AnalystAgent fires only when severity ≥ 5 or symptoms are detected (~\$0.015). A 'I'm fine' costs nothing meaningful.

Cross-Modal Synthesis

AnalystAgent fuses 60-day conversation history with Apple Watch biometrics (HR, SpO₂, HRV, sleep stages). Divergence between reported status and vitals is itself the alert — e.g. 'I feel fine' + SpO₂ 93%.

Hybrid RAG (FAISS + BM25)

Responses grounded in curated geriatric knowledge via Reciprocal Rank Fusion. Eliminates medical hallucinations and substantiates the \$29/mo clinical trust premium.

Forced Tool Calling

log_health_entry is force-called before every warm reply — guaranteeing qualitative conversation converts into structured DB rows powering every downstream alert.